Lab 5: Inference for a Single Mean: One Sample t-test

Part 1: Scenario & Overview

In this lab, we will explore the use of the t-test to compare a sample mean to a known population mean. We will use SPSS to perform the analyses and interpret the results.

Here is the research question we will be investigating:

> Research Question
>
> Does the new weight loss program with financial incentives result in a significantly different average weight loss over a 4-month period compared to the standard population average of 10 pounds?

Here is the scenario to that we will be working with:

A fitness center wants to determine if their new weight loss program is effective. It is believed that the average weight loss for individuals who participate in a standard weight loss programs is 10 pounds over a 4-month period. The fitness center wants to see if their program, which includes financial incentives, leads to a different average weight loss than this known population mean (10 lbs).

Study Design: The researchers recruited 38 individuals who wanted to lose weight and randomly assigned them to either an incentive group (given financial incentives) or a control group (not given any financial incentives). After 4 months, the researchers recorded the amount of weight lost (before - after) by each individual. Three months after the initial 4 months, they also recorded the weight lost by each individual again to see if the weight loss was maintained. However, financial incentives were only given during the first 4 months.

Learning Objectives

By the end of this lab, you should be able to:

- State null and alternative hypotheses for a population mean.
- Perform a one-sample t-test in SPSS.
- Interpret the results of a one-sample t-test in the context of a research question.

Stating Hypotheses

Null Hypothesis: The average weight loss for individuals in the incentive group is still equal to 10 pounds.

In symbols: $H_0 : \mu = 10$

Alternative Hypothesis: The average weight loss for individuals in the incentive group is different from 10 pounds.

In symbols: $H_1 : \mu \neq 10$

Checking Assumptions

There are two main assumptions to consider before running a one-sample t-test:

- Independence: The individuals in the sample are independent of each other. This means that there is no predictable relationship between one individual and another. This is generally satisfied if the data were collected through random sampling or random assignment. The data in this lab were collected through random assignment, so we can assume that the independence assumption is satisfied.
- Normality: The sampled population must be normal. This assumption is necessary part of the math underlying t-tests. Violating this assumption has little practical impact especially if the sample size is sufficiently large (often at least 30). Our sample size is 38, which is sufficiently large to satisfy this assumption.

Part 2: The Data

Importing Data into SPSS

The data set is available in the following link. Download the data.

Weight Loss Data.

After downloading the dataset, import it into SPSS, using the following steps:

File > Import Data > CSV Data..., browse to the file location, and click **OK** on the import wizard (accept the default settings).

Make sure the measurement scales are set as follows `WeightLoss` = Scale, `Group` = Nominal, `Month7Loss` = Scale.

See the screenshot below for reference:

	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Ro
1	WeightLoss	Numeric	4	1		None	None	8	Right	Scale	Inpu
2	Group	String	9	0		None	None	9	Left	Nominal	Inpu
3	Month7Loss	Numeric	4	1		None	None	8	Right	Scale	Inpu

variable Description

The variable description is as follows:

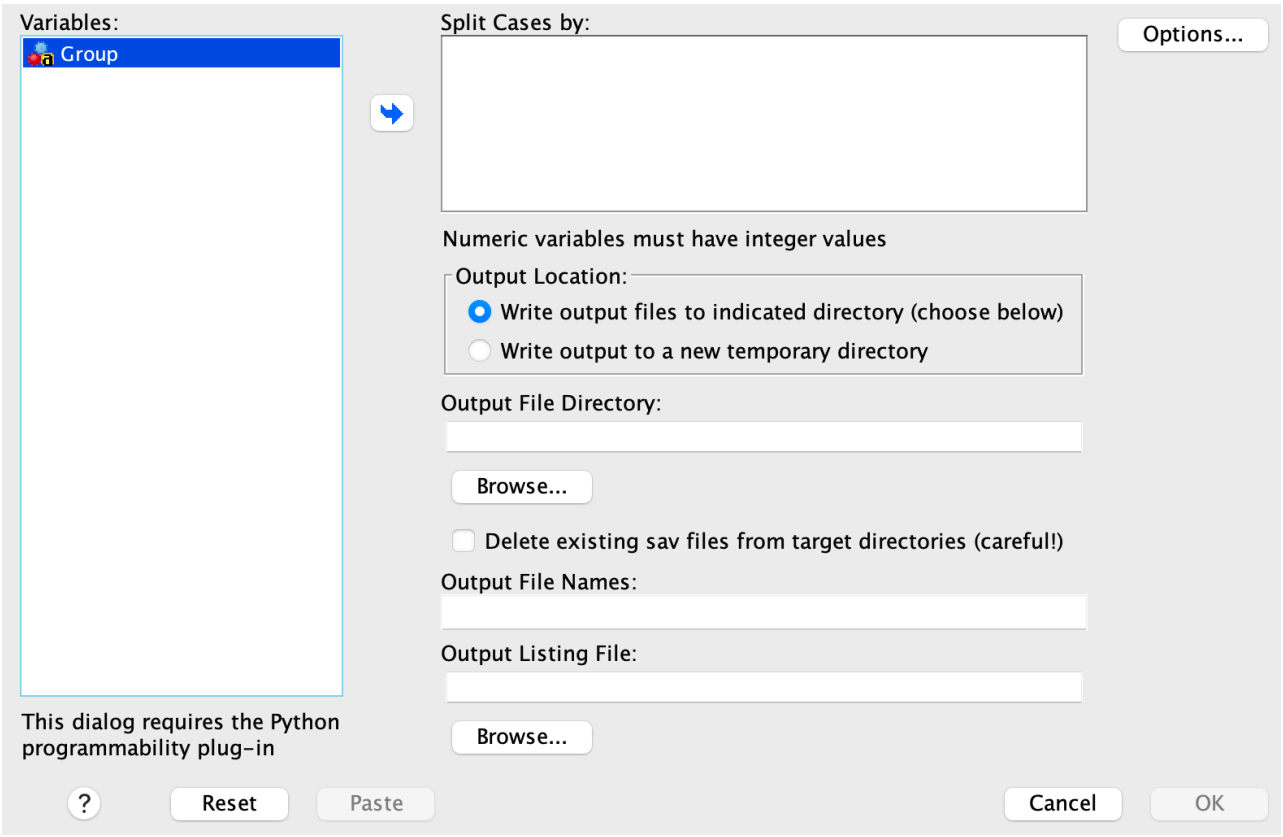
- ***Group*** = Whether someone was in the Incentive or Control group.- WeightLoss = The amount of weight lost by each individual
- ***Weightloss*** = The amount of weight lost by each individual at the 4-month follow-up.
- ***Month7Loss*** = The amount of weight lost by each individual at the 7-month follow-up

Click on data view to confirm that you have 38 individuals in the data set. The `WeightLoss` variable is calculated as the difference between the weight before and after the 4-month period. Negative values (e.g., case 14) indicate that the individual gained weight.

Data Filtering

We are only interested in the individuals in the incentive group for this lab. One way to isolate the incentive group is to use the **Split into files** feature in SPSS. Here are the steps:

Data > Split into Files. The following window will pop up:



Send the `Group` variable into the **Split Cases by** box, then click on **Browse** to specify the file location then click **OK**.

Next, navigate to the file location in your computer and open the file named `Incentive.sav`. This file contains members of the incentive group and their weight loss variables. This is the data we will use for the one-sample t-test.

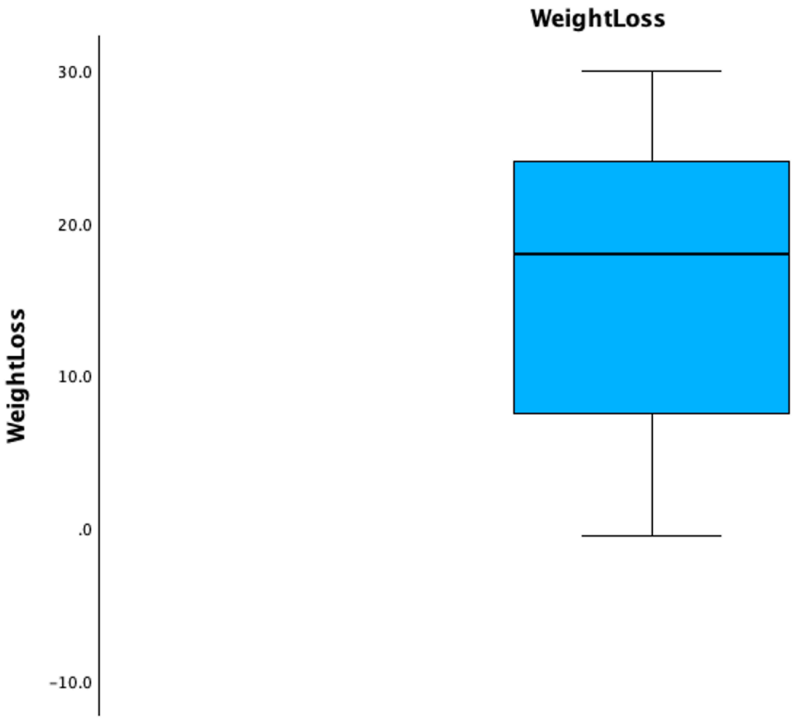
Part 3: Exploratory Analysis

Before running the t-test, we will perform some exploratory analysis to understand the data better. We can start by exploring the spread of the data using a box plot:

Run the following steps:

Graphs > Chart Builder > Box plot. Move `WeightLoss` into the **Y-axis** box. Click **OK**.

The box plot as shown below should appear in the output window:



Checkpoint

Based on the box plot, how would you describe the spread and shape of the `WeightLoss` variable for the incentive group? Are there any suspected outliers?

Checkpoint

Generate a histogram to visualize the distribution of the `WeightLoss` variable. What characteristics of the distribution are visible in the histogram that cannot be easily detected in the box plot?

Checkpoint

Calculate summary statistics (e.g., mean, etc) to examine the central tendency and spread of the `WeightLoss` variable.

Part 4: Running the t-test

To run the one-sample t-test, follow these steps:

- Analysis > Compare Means and Proportions > One-Sample T Test.
- Move `WeightLoss` variable into the **Test Variable(s)** box.
- Enter `15` in the **Test Value** box. Make sure the "Estimate effect size" check box is checked.
- Click on **Options** and check that the confidence interval is set to 95%. Click **Continue**.
- Click **OK**.

The following tables should pop up in the output window:

One-Sample Statistics

	N	Mean	Std. Deviation	Std. Error Mean
WeightLoss	17	15.676	9.4140	2.2832

One-Sample Test

	Test Value = 10						
			Significance		Mean	95% Confidence Interval of t Difference	
	t	df	One-Sided p	Two-Sided p	Difference	Lower	Upper
WeightLoss	2.486	16	.012	.024	5.6765	.836	10.5

The output shows both summary statistics and the results of the one-sample t-test. The summary statistics include the mean, standard deviation, and standard error. The t-test results (second table) include the t-statistic, degrees of freedom, p-value (one-sided and two-sided), and the confidence interval.

The P-value measures the chance of obtaining an average weight loss of 15 lbs or more extreme if the null hypothesis (average = 10 lbs) is true. Our obtained P-value is extremely small (less than the significance level of 0.05), so we reject the null hypothesis and conclude that the average weight loss for individuals in the incentive group is significantly different (higher) from 10 pounds.

Part 5: Output Interpretation

The output above provides you with the t-statistic, degrees of freedom, p-value (one-sided and two-sided), and the confidence interval, among others.

In scientific literature, you would see the one sample t-test results above reported in a manner similar to the following:

> APA-style Write-up
>
> The participants had an average weight loss of $M = 15.68$ lbs with $SD = 9.41$ lbs over the four-month period during which they were offered financial incentives. Statistical analysis indicates that the mean weight loss was significantly higher than population mean of 10 lbs, $t(17) = 2.486$, $p = .024$, $d = .603$, 95% CI [.836, 10.517].

Please see end of chapter 9 in the textbook for more details and examples on APA-style reporting of t-test results.

NOTE: The confidence interval is interpreted as follows: We are 95% confident that the true mean weight loss for individuals in the incentive group is between .836 and 10.517 pounds. It is strongly recommended that results of hypothesis tests be reported with confidence intervals when possible. Confidence intervals provide more information than a simple p-value and are more informative for decision-making.

Take-Home Lab Exercises

In today's lab exercise, you will perform a one-sample t-test to determine if the average weight loss at the seventh month for individuals in the incentive group is significantly different from 10 lbs. Recall that the financial incentives were only given during the first four months, so if the weight loss is not maintained, we would have further evidence that the financial incentives were effective in the short term but not in the long term.

Question 1

Create a box plot to visualize the weight loss 3 months after the initial 4 months (`Month7Loss`). How would you describe the spread and shape of the `Month7Loss` variable for the incentive group? Are there any suspected outliers?

Question 2

Create a histogram to visualize the distribution of the `Month7Loss` variable. What characteristics of the distribution are visible in the histogram that cannot be easily detected in the box plot?

Question 3

Run a one-sample t-test to determine if the average weight loss at the seventh month for individuals in the incentive group is significantly different from 10 lbs. Look at the output and answer the following questions:

a) What is the P-value for the two-sided test? What does this mean in the context of the research question?

b) What is the 95% confidence interval for the mean weight loss at the seventh month? How would you interpret this confidence interval in the context of the research question?

c) What is the effect size (Cohen's d) for this test? How would you interpret this effect size in the context of the research question?

Question 4

Write an APA-style interpretation of the results of the one-sample t-test for the `Month7Loss` variable. Be sure to include the mean, standard deviation, t-statistic, degrees of freedom, p-value, effect size, and confidence interval in your interpretation.

Submission Checklist

☐ A saved SPSS data (.sav) file showing the variables and data (includes $z - scores$).
☐ SPSS output file (.spv) showing the analyses performed.
☐ Completed responses for the Questions above. This document should be typed (Times New Roman, Font size 12, double spaced). The document should include necessary SPSS output tables and graphs. Remember that SPSS output tables and graphs can be copied and pasted into a word document.